TG: @Chalnaayaaar

WAVES

1. Motion of air from one place to the other is known as -

(1) Sound wave

(2) Transverse wave

(3) Wind

(4) SHM

2. Which of the following wave not required medium -

(1) Seismic wave

(2) Water waves

(3) Sound waves

(4) X - rays

3. Which property of the medium helps to generate a mechanical wave in a medium or helps in energy transfer :-

(1) elastic forces

(2) inertia

(3) both (1) & (2)

(4) none

4. Consider the following statements about sound passing through a gas.

(A) The pressure of the gas at a point oscillate in time

(B) The position of a small layer of the gas oscillate in time

(1) Both A & B are correct

(2) A is correct but B is wrong

(3) B is correct but A is wrong

(4) Both A & B are wrong

5. Which of the following wave can generate in spring.

(1) Only Longitudinal

(2) Only transverse

(3) Both longitudinal & transverse

(4) None

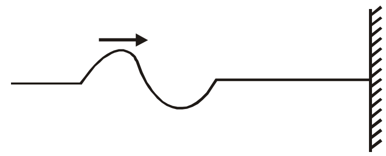
6. Which relation is correct for speed of sound wave in solid (s), liquid(ℓ) & Gases(g) :-

(1) $V_S > V_\ell > V_g$

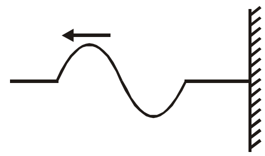
(2) $V_S < V_\ell < V_g$

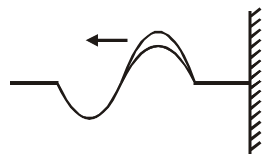
(3) $V_s = V_\ell = V_g$

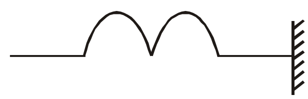
(4) $V_\ell > V_g > V_s$

7. 

Which diagram represent the correct reflected wave.

(1) 

(2) 

(3) 

(4) 

8. Select the correct statements :-

(1) In standing wave amplitude is fixed at a given location but, it is different at different locations

(2) All particles vibrate in same phase in between successive nodes

(3) A stationary wave pattern resulting from super position of two travelling wave in opposite direction (same frequency waves)

(4) All the above

9. In which of the following wave Doppler shifts are same irrespective of whether the source moves or the observer moves towards each other.

(1) Ultrasonic (2) Infrasonic

(3) Infrared (4) Audible

10. Doppler effect used in -

(1) sonography

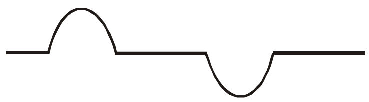
(2) echocardiogram

(3) astrophysics, military areas

(4) all the above

TG: @Chalnaayaaar

11. When both wave superimpose completely then the energy is in the form of -



(1) Kinetic Energy

(2) Potential Energy

(3) Partial potential & partial K.E

(4) Energy will completelty Loss

12. In a harmonic plane progressive wave which factor is different for all the particles in a wave.

(1) Amplitude (2) Frequency

(3) Phase (4) Time period

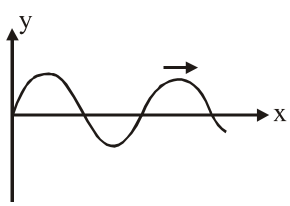
13. Relative to an observer at rest in a medium the speed of a mechanical wave in that medium doesn't depends on -

(1) Elasticity of medium (2) Mass of medium

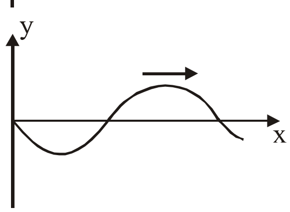
(3) Density of medium (4) Vel. of the source

14. Which wave represent the equation of wave $y = A \sin(kx - \omega t)$:-

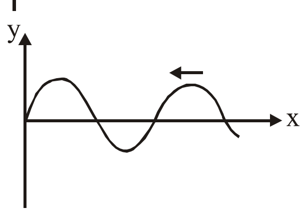
(1)



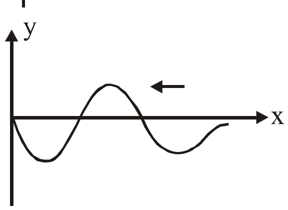
(2)



(3)



(4)



15. A narrow sound pulse is sent across a medium. The pulse have a definite -

(1) Frequency (2) Speed

(3) Wave Length (4) Time Period

16. Which of the wave can't propagate in vacuum:-

(1) Radiowave (2) Seismic wave

(3) X - rays (4) Light wave

17. Which of the wave is non-mechanical waves in nature.

(1) Wave on a string (2) Water waves

(3) X - Rays (4) Sound waves

18. Wave propagating on a streched string is:-

(1) Electromagnetic (2) Longitudinal

(3) Transverse (4) Non-Mechanical

19. Sound waves generated in a pipe filled with air by moving the piston up and down are :-

(1) Electromagnetic (2) Longitudinal

(3) Transverse (4) Non-mechanical

20. In the medium steel which type of mechanical wave can propagate :-

(1) Only Transverse

(2) Only Longitudinal

(3) Both Transverse and Longitudinal

(4) None of the above

21. Which one is the displacement function for a longitudinal wave. (Where S is the displacement of a medium particle :-

(1) $S(x_1, t) = a \sin(\omega t - ky + \phi)$

(2) $S(x_1, t) = a \sin(\omega t - kx + \phi)$

(3) $S(y, t) = a \sin(\omega t - kx + \phi)$

(4) $S(x, t) = a \sin(\omega t - kz + \phi)$

22. A wave travelling along a string is described by $y(x, t) = 0.005 \sin(20\pi t - 10\pi x + \frac{\pi}{2})$ cm where x is in cm and t is in sec. Calculate the displacement y of the wave at a distance x = 1 cm and time t = 1 sec.?

(1) $y = 0.0025$ cm

(2) $y = -0.0025$ cm

(3) $y = 0.005$ cm

(4) $y = -0.005$ cm

23. Speed of sound in air at room temperature is approximately 343 m/s. What can be the speed of sound in Hydrogen at room temperature approximately.

(1) 1284 m/s (2) 2089 m/s

(3) 686 m/s (4) 440 m/s

Z:\NODE02\B0B0-BA\NCERT BASED OBJECTIVE_PHY(INSIDER)\ENG\XI\14. WAVES.P65

TG: @Chalnaayaaar

24. Find out correct statement :-

(1) A travelling wave or pulse suffers a phase change of $\frac{\pi}{2}$ on reflection at a rigid boundary

(2) A travelling wave or pulse suffers no phase change on reflection at an open boundary

(3) A travelling wave suffers same phase change from rigid and open boundary

(4) None of these

25. A pipe, 30 cm long, is open at both ends. which harmonic mode of the pipe resonates a 1.1 kHz source? will resonance with the same source be observed if one end of the pipe is closed? Take the speed of sound in air as 330 m/s.

(1) 4^{th} Harmonic, yes

(2) II Harmonic, yes

(3) 4th Harmonic, No

(4) II Harmonic, No

26. Two sitar strings A and B playing the note 'Dha' are slightly out of tune and produces beats of frequency 5Hz. The tension of the string B is slightly increased and the beat frequency is found to decrease to 3Hz. What is the original frequency of B if the frequency of A is 427 Hz?

(1) 422 Hz (2) 432 Hz

(3) 424 Hz (4) 430 Hz

27. Find incorrect statement :-

(1) In a stationary wave, wavelength λ is twice the distance between two consecutive nodes

(2) In a progressive wave, wavelength λ is distance between two consecutive points of same phase at a given time.

(3) If k is angular wave number, wavelength λ is $\frac{\pi}{k}$.

(4) Frequency n of a wave is defined as $\frac{1}{T}$ and is related to angular frequency by $n = \frac{\omega}{2\pi}$

28. A pipe of length L with one end closed and other end open vibrates with frequencies given by :-

(1) $f = n\left(\frac{v}{2L}\right)$, n = 0, 1, 2, 3, ...

(2) $f = \left(n+\frac{1}{2}\right)\left(\frac{v}{2L}\right)$, n = 0, 1, 2, 3, 4 ...

(3) $f = (2n + 1)\frac{v}{2L}$, n = 1, 3, 5 ......

(4) $f = \frac{nv}{4L}$, n = 2, 4, 6 ......

29. Waves produced by a motorboat sailing in water :-

(1) Only transverse

(2) Only Longitudinal

(3) Transverse and Longitudinal

(4) None of these

30. Which of following waves are governed by Newton's laws of motion :-

(1) Mechanical waves

(2) Non-Mechanical waves

(3) Matter waves

(4) None of these

31. The picture of a progressive transverse wave at a particular instant of time gives -

(1) Shape of wave

(2) Motion of the particle of medium

(3) Velocity of wave

(4) None of above

32. The displacement of the wave given by equation $y(x, t) = a\sin(kx - \omega t + \phi)$ where $\phi = 0$ at point x and t = 0 is same as that at point:-

(1) $x + 2n\pi$ (2) $x + \frac{2n\pi}{k}$

(3) $kx + 2n\pi$ (4) None

33. Nature of ocean waves is :-

(1) Longitudinal (2) Transverse

(3) Both (4) None

TG: @Chalnaayaaar

34. The waves on the surface of water are called as -
(1) P - wave (2) Shear wave
(3) Both (4) None

35. If two sounds of very close frequencies 256 Hz and 260 Hz reach our ear simultaneously, then we hear a sound of frequency -
(1) 258 Hz (2) 4 Hz
(3) 2Hz (4) None

36. Let a wave $y(x, t) = a\sin(kx - \omega t)$ is reflected from an open boundary and then the incident and reflected waves overlaps then amplitude of resultant wave -
(1) $2a\sin(kx)$ (2) $2a\cos(kx)$
(3) $2a\sin\left(\frac{kx}{2}\right)$ (4) $a\sin(kx)$

37. When a high - pressure pulse of air travelling down an open pipe reaches the other end then
(1) High pressure pulse starts travelling up the pipe
(2) Low pressure pulse starts travelling up the pipe
(3) Normal pressure pulse starts travelling up the pipe
(4) None of these

38. During Propagation of a plane progressive mechanical wave, which of the following statement is incorrect:-
(1) All the particles are vibrating in the same phase
(2) Amplitude of all the particles is equal
(3) Particles of the medium executes SHM
(4) Wave velocity wavelength depends upon the nature of the medium

39. Which of the following statements are false for stationary wave :-
(1) All the particles cross their mean position at the same time
(2) All the particles are oscillating with same amplitudes
(3) There is no energy transfer across any plane
(4) There are some particles which are always at rest

40. When the observer moves towards stationary source then which of the following is true regarding frequency and wavelength of wave observed by the observer -
(1) More frequency, less wavelength
(2) More frequency, more wavelength
(3) Less frequency, more wavelength
(4) More frequency, constant wavelength

41. In a sound wave, a displacement node is a pressure - antinode because -
(1) Pressure and density are maximum at displacement node
(2) Pressure and density are minimum at displacement node
(3) Change in pressure and change in density are maximum at displacement node
(4) None

42. A meter long tube (open at one end) with a movable piston at the other end, shows resonance with a fixed frequency source (a tuning fork of frequency 340 Hz) when the tube length is 25.5 cm or 76.5 cm. Estimate the speed of sound in air (Ignore edge effect) -
(1) 347 m/s (2) 333 m/s
(3) 330 m/s (4) 370 m/s

43. The transverse stationary wave on a stretched wire (Clamped at its both ends) is given by -

$$y(x, t) = 0.06 \sin\left(\frac{2\pi}{3}x\right)\cos(120\pi t)$$

where x and y are in m and t in second. The length of string is 1.5 m and it's mass is 3×10^{-2}kg then all the points on the wire vibrate with -
(1) Same phase
(2) Same amplitude
(3) Same energy
(4) Different frequency

44. Speed of sound wave in air
(1) is independent of temperature
(2) increases with pressure
(3) increases with increase in humidity
(4) decreases with increase in humidity

TG: @Chalnaayaaar

45. Change in temperature of the medium changes

(1) frequency of sound waves

(2) amplitude of sound waves

(3) wavelength of sound waves

(4) loudness of sound waves

46. A sond wave is passing through air column in the form of compression and rarefaction. In consecutive compression and rarefactions.

(1) density remains constant

(2) Boyle's law is obeyed

(3) bulk molulus of air oscillates

(4) there is no transfer of heat

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**3**	**4**	**3**	**1**	**3**	**1**	**1**	**4**	**3**	**4**	**1**	**3**	**4**	**1**	**2**
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	**2**	**3**	**3**	**2**	**3**	**2**	**3**	**1**	**2**	**4**	**1**	**3**	**2**	**3**	**1**
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	**1**	**2**	**3**	**3**	**1**	**2**	**2**	**1**	**2**	**4**	**3**	**1**	**1**	**3**	**3**
Que.	46														
Ans.	**4**														